EXPERIMENT NO. 5

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Input Code:

```
#include <stdio.h>
#define MAX 5
int stack[MAX];
int top = -1;
void push() {
   int val;
   if (top == MAX - 1) {
      printf("\nStack Overflow! Cannot push more elements.\n");
      return;
   }
   printf("Enter value to push: ");
   scanf("%d", &val);
   top++;
   stack[top] = val;
   printf("Successfully pushed %d into the stack.\n", val);
}
void pop() {
   if (top == -1) {
      printf("\nStack Underflow! Stack is empty.\n");
      return;
   }
   printf("\nPopped element: %d\n", stack[top]);
   top--;
}
void display() {
   if (top == -1) {
```

```
        printf("\nStack is empty!\n");
        return;
    }
    printf("\n--- Stack Contents (Top to Bottom) ---\n");
    for (int i = top; i >= 0; i--) {
        printf("| %d |\n", stack[i]);
    }
    printf("-------+\n");
    printf("Current TOP Position (Index): %d\n", top);
    printf("Current TOP Element Value: %d\n", stack[top]);
}
int main() {
    int choice;
    do {
        printf("-----STACK OPERATIONS (ARRAY)-----\n");
        printf("1. Push\n");
        printf("2. Pop\n");
        printf("3. Display Stack & Top Position\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                push();
                break;
            case 2:
                pop();
                break;
            case 3:
```

```
            display();
            break;
        case 4:
            printf("\nExiting program. Goodbye!\n");
            break;
        default:
            printf("\nInvalid choice! Please try again.\n");
    }
  } while (choice != 4);
  return 0;
}
```

Output:

```
-----STACK OPERATIONS (ARRAY)-----
1. Push
2. Pop
3. Display Stack & Top Position
4. Exit
Enter your choice: 1
Enter value to push: 12
Successfully pushed 12 into the stack.
-----STACK OPERATIONS (ARRAY)-----
1. Push
2. Pop
3. Display Stack & Top Position
4. Exit
Enter your choice: 1
Enter value to push: 13
Successfully pushed 13 into the stack.
-----STACK OPERATIONS (ARRAY)-----
1. Push
2. Pop
3. Display Stack & Top Position
4. Exit
Enter your choice: 3

--- Stack Contents (Top to Bottom) ---
| 13 |
| 12 |
-------+
Current TOP Position (Index): 1
Current TOP Element Value: 13
```

```
-----STACK OPERATIONS (ARRAY)-----
1. Push
2. Pop
3. Display Stack & Top Position
4. Exit
Enter your choice: 2

Popped element: 13
-----STACK OPERATIONS (ARRAY)-----
1. Push
2. Pop
3. Display Stack & Top Position
4. Exit
Enter your choice: 3

--- Stack Contents (Top to Bottom) ---
| 12 |
-------+
Current TOP Position (Index): 0
Current TOP Element Value: 12
-----STACK OPERATIONS (ARRAY)-----
1. Push
2. Pop
3. Display Stack & Top Position
4. Exit
Enter your choice: 4

Exiting program. Goodbye!
```